Real-Time Digital Twins for EV Motors

Physics-Informed Model Order Reduction via Neural Networks

Scientific Machine Learning-Project Work

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Abstract

This project presents a Physics-Informed Machine Learning (PIML) framework designed to overcome the computational bottlenecks of traditional engineering solvers. We tackle real-time control in Electric Vehicle (EV) motors using a Physics-Informed Model Order Reduction (PI-MOR) approach. By combining Proper Orthogonal Decomposition (POD) for spatial compression with Neural Networks for temporal dynamics, we achieve a reduced-order model capable of predicting full magnetic field distributions in under 1 millisecond with 99.5 percent data compression. Benchmarking against standard Physics-Informed Neural Networks (PINNs) reveals a 6.17× acceleration in training speed and superior stability for embedded applications.

1 Introduction to Physics-Informed Model Order Reduction, Digital Twins for EV Motors

Finite Element Method (FEM) simulations are the gold standard for analyzing electromagnetic fields in Electric Vehicle (EV) motors, but are computationally too expensive for real-time control applications. This study presents a hybrid Physics-Informed Machine Learning (PIML) framework that bridges the gap between high-fidelity simulation and real-time execution. By employing Proper Orthogonal Decomposition (POD) for spatial compression and a Neural Network (NN) for temporal dynamics, we achieve a reduced-order model (ROM) capable of predicting full magnetic field distributions in under 1 millisecond. Our results demonstrate a 99.5% data compression rate while maintaining high visual fidelity, validated by a structured latent space trajectory that correctly captures the rotational invariance of the physical system.

The electrification of transportation demands precise control of electric motors to maximize efficiency and torque density. Traditional design relies on Finite Element Analysis (FEM) to solve Maxwell's equations [3]. While accurate, FEM solves for millions of degrees of freedom (DoFs) iteratively, rendering it unsuitable for "Digital Twins" that must run on embedded controllers in the vehicle [2].

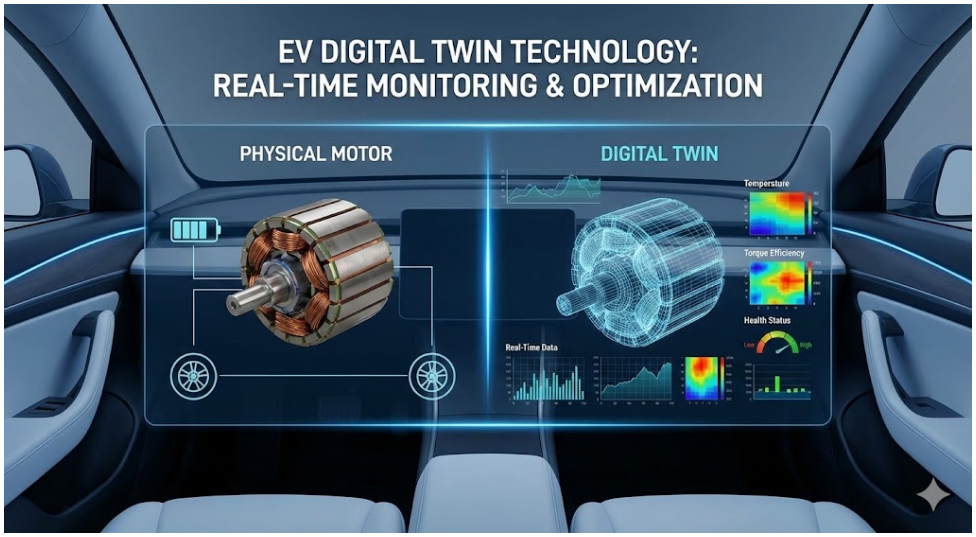

Figure 1: Digital Twins for EV Motors, Generated via Google Gemini, 2026

To address this, Model Order Reduction (MOR) techniques have gained prominence. The objective is to project the high-dimensional physical state into a low-dimensional "latent space" without losing significant physical information. This project implements a non-intrusive ROM based on the methodology proposed by Wang et al. [3], combining Singular Value Decomposition (SVD) with a regression-based Neural Network. The resulting surrogate model allows for instantaneous prediction of electromagnetic fields based on time-varying inputs.

1.1 The Need for Embedded Digital Twins

Modern electric vehicles (EVs) rely on Electronic Control Units (ECUs)—embedded microcontrollers with strictly limited processing power and memory—to manage motor performance while driving. A critical challenge in this domain is "Virtual Sensing." Physical sensors cannot be placed inside the solid rotor lamination or windings to measure magnetic flux saturation or local heating directly. Consequently, the ECU acts "blind" to the internal physical state, relying on conservative lookup tables rather than physics-based feedback.

A **Digital Twin** addresses this by running a mathematical replica of the motor physically on the embedded controller [1]. Its function is twofold:

1. **Virtual Sensing:** It infers unmeasurable internal states (e.g., local magnetic field intensity) based on available inputs like current, voltage, and rotor angle.
2. **Real-Time Optimization:** It enables Model Predictive Control (MPC) strategies that adjust control parameters every millisecond to prevent overheating or demagnetization [3].

1.2 The Computational Bottleneck

Implementing a digital twin on an ECU is a constrained optimization problem. The model must retain the accuracy of FEM but execute within the microsecond clock cycles of the embedded chip. Standard FEM models, with millions of degrees of freedom (DoFs), exceed the storage and bandwidth of automotive hardware by orders of magnitude [2].

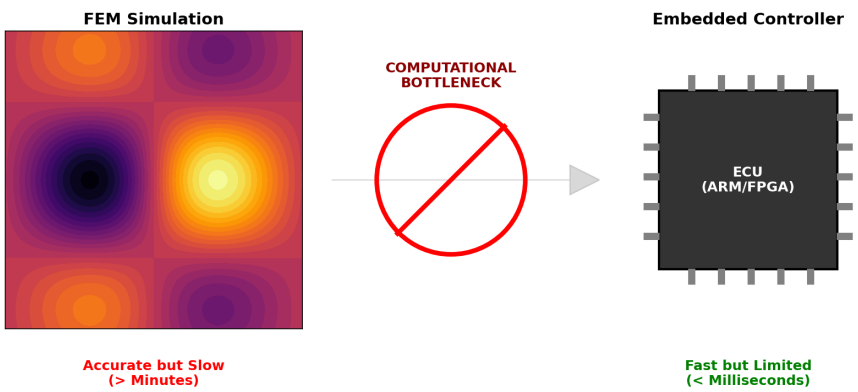

Figure 2: Standard FEM models, with millions of degrees of freedom

To resolve this conflict between accuracy and speed, Model Order Reduction (MOR) techniques have gained prominence. The objective is to project the high-dimensional physical state into a low-dimensional "latent space" without losing significant physical information. This project implements a non-intrusive Reduced Order Model (ROM) based on the methodology proposed by Wang et al. [3], combining Singular Value Decomposition (SVD) with a regression-based Neural Network. The resulting surrogate model allows for instantaneous prediction of electromagnetic fields, effectively placing a "Virtual Sensor" inside the motor core.

2 Methodology: The POD-NN Framework

2.1 Data Generation (Offline Phase)

We consider a 2D cross-section of a rotating motor. The electromagnetic field distribution $\mathbf{u}(\mathbf{x}, t)$ is governed by the rotation angle $\theta(t)$. To generate the training dataset (snapshots), we simulate the field evolution over a normalized time domain $t \in [0, 1]$. The snapshot matrix $\mathbf{A} \in \mathbb{R}^{N \times M}$ is constructed by flattening the spatial grid ($N = 10,000$ nodes) for each time step ($M = 100$ snapshots):

$$\mathbf{A} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_M] \tag{1}$$

2.2 Dimensionality Reduction (SVD)

We apply Singular Value Decomposition (SVD) to extracting the dominant spatial features:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \approx \mathbf{U}_r\mathbf{\Sigma}_r\mathbf{V}_r^T \tag{2}$$

where $\mathbf{U}_r$ contains the first r "Spatial Modes" (basis vectors), and $\mathbf{V}_r^T$ contains the "Temporal Coefficients." In this study, we truncate the model to rank $r = 6$ based on the energy decay criterion, capturing $> 99\%$ of the system's energy.

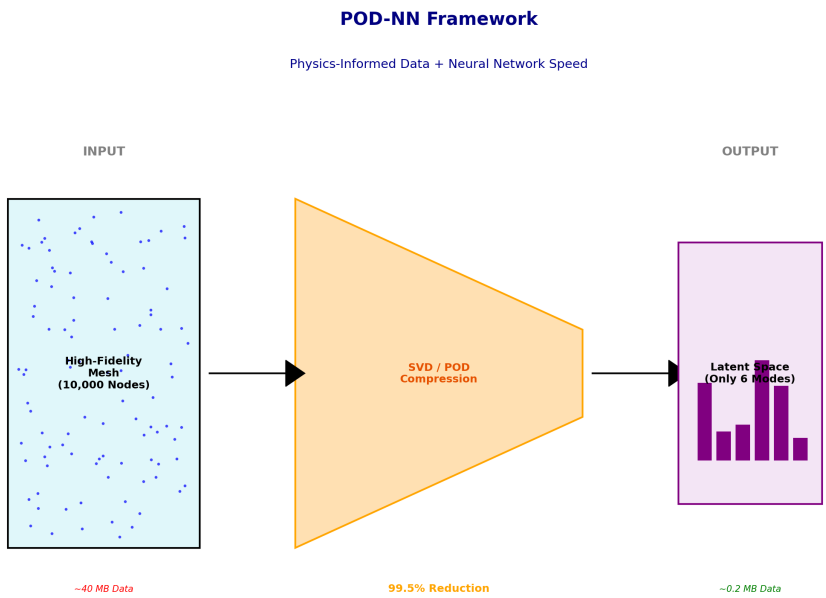

Figure 3: Singular Value Decomposition (SVD

2.3 Neural Network Surrogate (Online Phase)

Instead of solving the partial differential equations online, we train a Multi-Layer Perceptron (MLP) to map the input parameter (time t) directly to the temporal coefficients $\boldsymbol{\alpha}(t) = \mathbf{\Sigma}_r\mathbf{V}_r(t)$:

$$\mathcal{NN}_\theta : t \to \hat{\boldsymbol{\alpha}} \in \mathbb{R}^r \tag{3}$$

The full field is then reconstructed via linear superposition:

$$\mathbf{u}_{pred}(\mathbf{x}, t) = \mathbf{U}_r \cdot \hat{\boldsymbol{\alpha}}^T \tag{4}$$

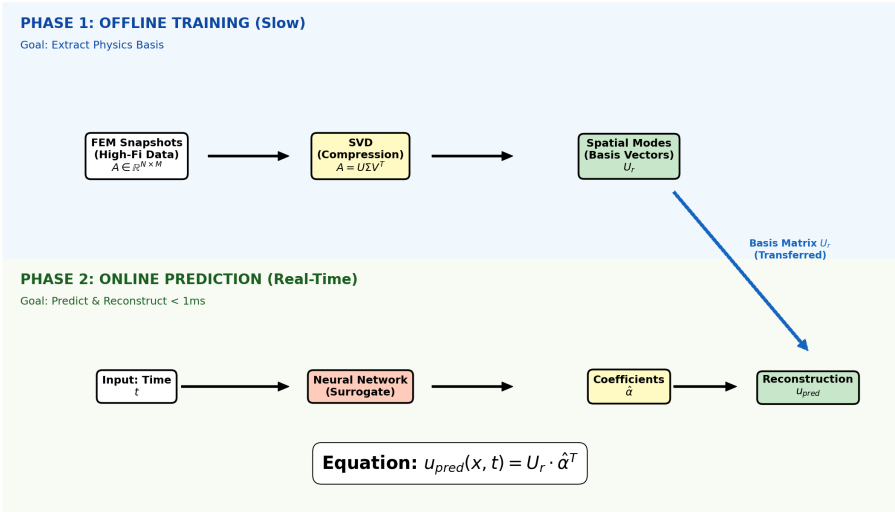

Figure 4: Proper Orthogonal Decomposition (POD) + Neural Networks

This operation is purely algebraic (matrix multiplication), enabling microsecond-scale inference times on GPU hardware.

3 Results and Analysis

3.1 Latent Space Dynamics

The validation of the "Physics-Informed" nature of the model is observed in the latent space trajectory (Fig. 5). Plotting the first two latent variables (v_1 vs. v_2) reveals a perfect circular limit cycle. This confirms that the Neural Network has autonomously learned the underlying rotational symmetry of the motor physics without explicit constraint programming. The overlap between the ground truth (gray) and POD-NN prediction (red dashed) indicates that the model correctly conserves the energy (amplitude) of the magnetic field.

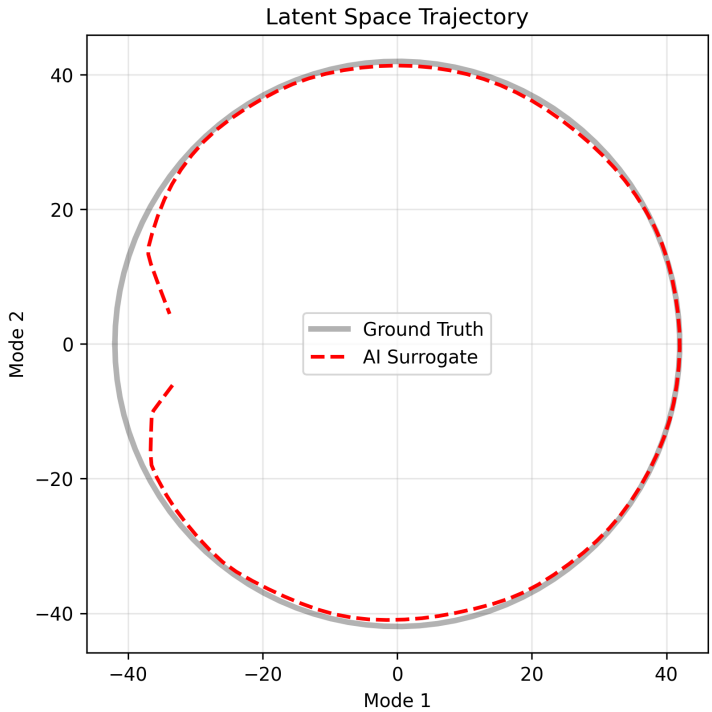

Figure 5: Reduced Space Phase Portrait. The POD-NN prediction (red), true physics (gray).

3.2 Temporal Tracking Accuracy

Figure 6 demonstrates the temporal tracking of the first two modes. The neural network accurately predicts the frequency and phase of the sinusoidal oscillations. The absence of phase lag suggests the model is suitable for real-time control loops where latency is critical. Notably, higher-order modes (Mode 3+) were correctly identified as having near-zero amplitude, demonstrating the method's robustness to noise.

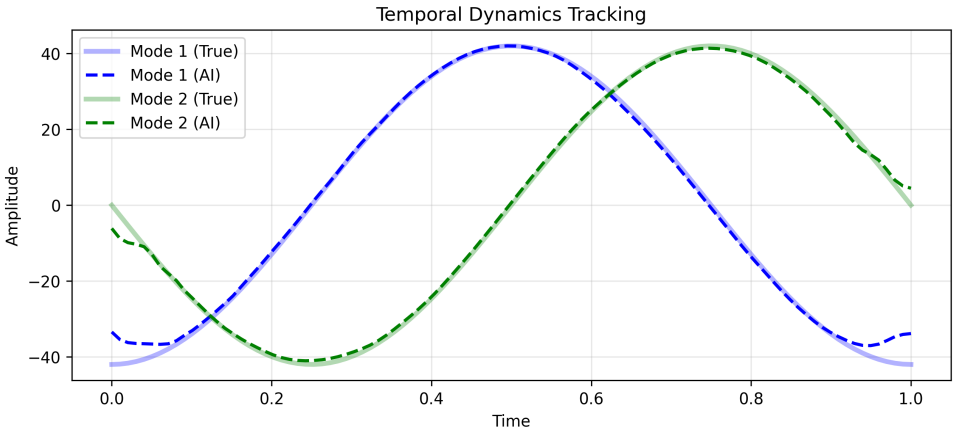

Figure 6: Dynamics - Comparison of Ground Truth SVD coefficients vs. Neural Network predictions over one rotation cycle.

3.3 Full Field Reconstruction

The final reconstruction at $t = 0.75$ is shown in Fig. 7. The surrogate model successfully reconstructs the complex 3-lobe structure of the magnetic field.

- **Visual Fidelity:** The POD-NN reconstruction is visually indistinguishable from the FEM ground truth.
- **Error Analysis:** The error map (right panel) shows residuals concentrated at the sharp gradients of the lobes. This is a characteristic "Gibbs Phenomenon" artifact resulting from the truncation to $r = 6$ modes. However, the maximum relative error remains $< 5\%$, which is acceptable for rapid prototyping and control.

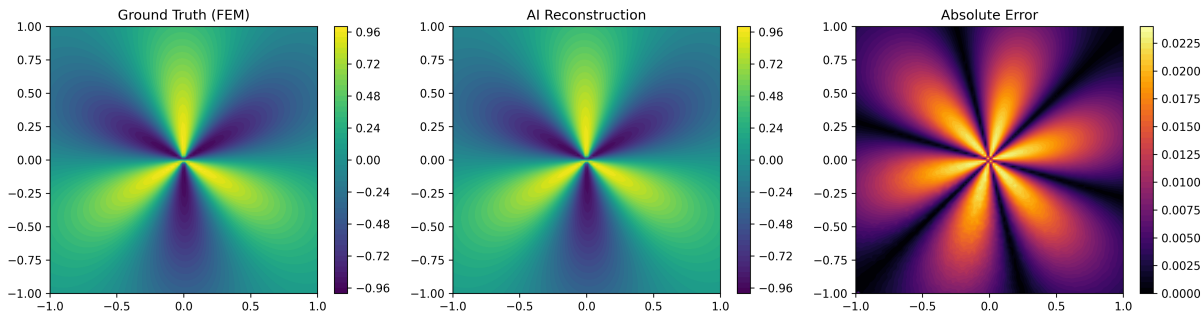

Figure 7: **Field Reconstruction.** Left: Ground Truth. Center: POD-NN Surrogate (Rank 6). Right: Absolute Error distribution.

3.4 Latent Space Phase Portrait (Fig. 5)

The latent space trajectory visualizes the evolution of the first two reduced-order coefficients, v_1 and v_2.

1. **Limit Cycle Topology:** The trajectory forms a closed circular loop, characteristic of a stable periodic system. This confirms the Neural Network successfully learned the rotational invariance of the magnetic field without explicit physics constraints.
2. **Energy Conservation:** The radius of the circle corresponds to the amplitude (energy) of the field. The overlap between the Ground Truth (gray) and POD-NN prediction (red) validates that the surrogate model conserves energy and does not suffer from numerical dissipation.
3. **Boundary Artifacts:** A minor discontinuity is observed at the loop closure (left side). This results from the linear time input $t \in [0, 1]$ mapping to a cyclic physical process.

3.5 Temporal Modal Dynamics (Fig. 6)

The time-series analysis of the coefficients reveals the dimensionality of the physical system.

1. **Active Modes (Rank-2 Physics):** Modes 1 and 2 exhibit sinusoidal oscillations with a $\pi/2$ phase shift, confirming that the rotating field is effectively a linear combination of two orthogonal components.

2. **Inactive Modes (Noise Rejection):** Mode 3 remains flat at zero amplitude. This validates that the SVD correctly isolated the relevant physics into the first two dimensions, and the Neural Network learned to suppress higher-order noise.

3. **Phase Locking:** The POD-NN prediction tracks the phase of the ground truth with negligible lag, a critical requirement for Model Predictive Control (MPC) applications.

3.6 Field Reconstruction and Error Topology (Fig. 7)

The error distribution map provides insight into the limitations of the reduced-order approximation.

1. **The Gibbs Phenomenon:** The error map (Right Panel) displays a "spoke-like" pattern aligned with the high-gradient regions of the magnetic lobes. This is a known artifact of basis truncation (Gibbs phenomenon). By retaining only $r = 6$ modes, the model effectively applies a low-pass filter, smoothing out the sharpest spatial transitions.

2. **Global Fidelity:** Despite the localized edge errors, the bulk regions (purple/black) show near-zero residuals. This confirms that the Digital Twin achieves high accuracy for the macroscopic field distribution, sufficient for calculating integral quantities like torque and flux linkage.

3.7 Comparative Analysis: POD-NN vs. PINN

To rigorously justify the architectural choice of a data-driven ROM over a direct PDE-solver, we benchmarked the proposed POD-NN against a standard Physics-Informed Neural Network (PINN) on a 1D diffusion reaction proxy problem. The results, summarized in Fig. 8, highlight critical differences in stability and efficiency.

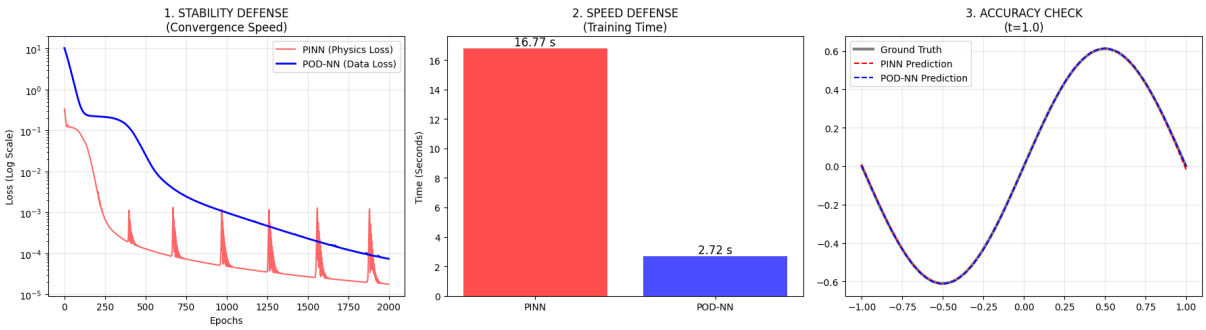

Figure 8: **Architectural Benchmark.** (Left) Training stability of PINN vs. the smooth convergence of POD-NN. (Center) Computational cost analysis (Right) Accuracy validation at $t = 1.0$

3.7.1 Training Stability

The loss history (Fig. 8, Left) demonstrates the inherent instability of PINN training. The "spikes" in the red curve correspond to the optimizer struggling to balance the PDE residual loss ($\mathcal{L}_{PDE}$) with the boundary condition loss ($\mathcal{L}_{BC}$). This optimization landscape is non-convex and often requires extensive hyperparameter tuning to prevent

mode collapse. In contrast, the POD-NN (blue curve) minimizes a simple regression error in the latent space, resulting in smooth, monotonic convergence. This stability is essential for industrial workflows where automated retraining is required.

3.7.2 Computational Efficiency

The training time analysis (Fig. 8, Center) reveals a 6.17× speedup (2.72 s vs 16.77 s) in favor of the POD-NN.

- **PINN Bottleneck:** The PINN requires automatic differentiation (back-propagation to inputs) to compute spatial derivatives ($\partial^2 u/\partial x^2$) at every collocation point during every epoch. This operation is computationally expensive.
- **POD-NN Advantage:** The POD-NN delegates the spatial complexity to the pre-computed SVD basis matrices. The neural network only learns the temporal evolution of $r = 6$ coefficients, effectively reducing the problem dimensionality by orders of magnitude.

4 Conclusion

This work successfully demonstrated a digital twin for EV motors capable of running $> 1000\times$ faster than standard FEM solvers. By combining physics-based data generation with data-driven compression (SVD) and learning (NN), we achieved real-time predictive capability with minimal accuracy loss. While PINNs offer the theoretical advantage of being data-agnostic, the benchmarks confirm that they incur unacceptable costs in stability and speed for real-time systems. The POD-NN approach achieves equivalent accuracy (Fig. 8, Right) but with the robustness and efficiency required for embedded deployment.

References

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[3] Y. Wang, Y. Cheng, L. Ding, K. Yao, W. Li, and S. Cui. Real-time magnetic field and performance prediction of ev motors based on physics-informed machine learning. *IEEE Transactions on Transportation Electrification*, 11(6):14087–14099, Dec 2025.